

Number System Conversions

Binary, Octal, Decimal, and Hexadecimal

Introduction

This assignment delves into the fascinating world of number system conversions. We will explore the four primary number systems – **Binary**, **Octal**, **Decimal**, and **Hexadecimal** – and learn how to convert numbers between them. Understanding these conversions is crucial in computer science and digital electronics, as different systems are used for different purposes. Let's embark on this journey to master all 12 possible conversions!

Number Systems Overview

Decimal (Base-10)

The number system we use daily. It has ten digits (0-9).

Binary (Base-2)

The language of computers. It has two digits (0 and 1).

Octal (Base-8)

Uses eight digits (0-7). Often used as a shorthand for binary.

Hexadecimal (Base-16)

Uses sixteen digits (0-9 and A-F). Commonly used to represent memory addresses and colors.

Conversion Techniques

We will now cover all 12 conversion processes, along with examples and practice exercises:

1. **Decimal to Binary**
2. **Decimal to Octal**
3. **Decimal to Hexadecimal**
4. **Binary to Decimal**
5. **Binary to Octal**
6. **Binary to Hexadecimal**
7. **Octal to Decimal**
8. **Octal to Binary**
9. **Octal to Hexadecimal**
10. **Hexadecimal to Decimal**

11. Hexadecimal to Binary

12. Hexadecimal to Octal

Decimal Conversions

1. Decimal to Binary

Method: Repeated division by 2. Record the remainders (0 or 1) at each step. The binary number is formed by reading the remainders from bottom to top.

Example: Convert 25 (decimal) to binary.

$$25 / 2 = 12, \text{ Remainder} = 1$$

$$12 / 2 = 6, \text{ Remainder} = 0$$

$$6 / 2 = 3, \text{ Remainder} = 0$$

$$3 / 2 = 1, \text{ Remainder} = 1$$

$$1 / 2 = 0, \text{ Remainder} = 1$$

Therefore, 25 (decimal) = 11001 (binary)

Exercise: Convert 42 (decimal) to binary.

2. Decimal to Octal

Method: Repeated division by 8. Record the remainders (0-7) at each step. The octal number is formed by reading the remainders from bottom to top.

Example: Convert 150 (decimal) to octal.

$$150 / 8 = 18, \text{ Remainder} = 6$$

$$18 / 8 = 2, \text{ Remainder} = 2$$

$$2 / 8 = 0, \text{ Remainder} = 2$$

Therefore, 150 (decimal) = 226 (octal)

Exercise: Convert 98 (decimal) to octal.

3. Decimal to Hexadecimal

Method: Repeated division by 16. Record the remainders (0-15). If the remainder is 10-15, represent it with the letters A-F (A=10, B=11, C=12, D=13, E=14, F=15). The hexadecimal number is formed by reading the remainders from bottom to top.

Example: Convert 450 (decimal) to hexadecimal.

$450 / 16 = 28$, Remainder = 2

$28 / 16 = 1$, Remainder = 12 (C)

$1 / 16 = 0$, Remainder = 1

Therefore, 450 (decimal) = 1C2 (hexadecimal)

Exercise: Convert 285 (decimal) to hexadecimal.

Binary Conversions

4. Binary to Decimal

Method: Multiply each digit by 2 raised to the power of its position (starting from 0 on the right) and sum the results.

Example: Convert 10110 (binary) to decimal.

$$(1 * 2^4) + (0 * 2^3) + (1 * 2^2) + (1 * 2^1) + (0 * 2^0) = 16 + 0 + 4 + 2 + 0 = 22$$

Therefore, 10110 (binary) = 22 (decimal)

Exercise: Convert 11011 (binary) to decimal.

5. Binary to Octal

Method: Group the binary digits into groups of three (from right to left). Add leading zeros if necessary. Convert each group of three into its octal equivalent.

Example: Convert 101101 (binary) to octal.

Grouped: 101 101

Octal equivalent: 5 5

Therefore, 101101 (binary) = 55 (octal)

Exercise: Convert 110011 (binary) to octal.

6. Binary to Hexadecimal

Method: Group the binary digits into groups of four (from right to left). Add leading zeros if necessary. Convert each group of four into its hexadecimal equivalent.

Example: Convert 1101011 (binary) to hexadecimal.

Grouped: 0110 1011

Hexadecimal equivalent: 6 B

Therefore, 1101011 (binary) = 6B (hexadecimal)

Exercise: Convert 10101010 (binary) to hexadecimal.

Octal Conversions

7. Octal to Decimal

Method: Multiply each digit by 8 raised to the power of its position (starting from 0 on the right) and sum the results.

Example: Convert 237 (octal) to decimal.

$$(2 * 8^2) + (3 * 8^1) + (7 * 8^0) = 128 + 24 + 7 = 159$$

Therefore, 237 (octal) = 159 (decimal)

Exercise: Convert 456 (octal) to decimal.

8. Octal to Binary

Method: Convert each octal digit to its 3-bit binary equivalent.

Example: Convert 72 (octal) to binary.

$$7 = 111$$

$$2 = 010$$

Therefore, 72 (octal) = 111010 (binary)

Exercise: Convert 53 (octal) to binary.

9. Octal to Hexadecimal

Method: First, convert the octal number to binary. Then, convert the binary number to hexadecimal.

Example: Convert 64 (octal) to hexadecimal.

$$64 \text{ (octal)} = 110100 \text{ (binary)}$$

$$110100 \text{ (binary)} = 34 \text{ (hexadecimal)}$$

Therefore, 64 (octal) = 34 (hexadecimal)

Exercise: Convert 25 (octal) to hexadecimal.

Hexadecimal Conversions

10. Hexadecimal to Decimal

Method: Multiply each digit by 16 raised to the power of its position (starting from 0 on the right) and sum the results. Remember that A=10, B=11, C=12, D=13, E=14, F=15.

Example: Convert 2A (hexadecimal) to decimal.

$$(2 * 16^1) + (10 * 16^0) = 32 + 10 = 42$$

Therefore, 2A (hexadecimal) = 42 (decimal)

Exercise: Convert 3F (hexadecimal) to decimal.

11. Hexadecimal to Binary

Method: Convert each hexadecimal digit to its 4-bit binary equivalent.

Example: Convert A5 (hexadecimal) to binary.

$$A = 1010$$

$$5 = 0101$$

Therefore, A5 (hexadecimal) = 10100101 (binary)

Exercise: Convert C8 (hexadecimal) to binary.

12. Hexadecimal to Octal

Method: First, convert the hexadecimal number to binary. Then, convert the binary number to octal.

Example: Convert 1A (hexadecimal) to octal.

$$1A \text{ (hexadecimal)} = 00011010 \text{ (binary)}$$

$$00011010 \text{ (binary)} = 032 \text{ (octal)}$$

Therefore, 1A (hexadecimal) = 32 (octal)

Exercise: Convert 2B (hexadecimal) to octal.

Conclusion

Congratulations! You have now explored all 12 number system conversions. Practice these conversions regularly to solidify your understanding. Mastery of these concepts is essential for anyone working with computers and digital systems. Good luck with the exercises!